EXPERIMENT:8

Aim: To implement CI/CD pipeline to host a website on AWS

Lab Outcome: LO3

LO3: Build and manage automated CI/CD pipelines for application development, testing, packaging, and deployment.

Theory:

Continuous Integration and Continuous Deployment (CI/CD)

Continuous Integration (CI) is a software development practice where developers merge their code changes frequently into a central shared repository, followed by automated builds and automated test executions to detect integration bugs early. Continuous Delivery/Deployment (CD) extends this practice by automating the release process so that verified code changes can be automatically packaged and deployed directly to staging or production environments without manual intervention.

Implementing CI/CD drastically shortens the software development lifecycle (SDLC), minimizes manual deployment errors, improves software stability, and enables rapid iteration of software products.

Key Components of AWS CI/CD Architecture

- Source Repository (GitHub): Acts as the centralized version control system (VCS). Webhooks or OAuth connection points alert the cloud pipeline instantaneously whenever commits are pushed to specific branches (e.g., main).
- AWS CodePipeline: A continuous delivery orchestration service that models, visualizes, and coordinates every step of the release process. It manages the

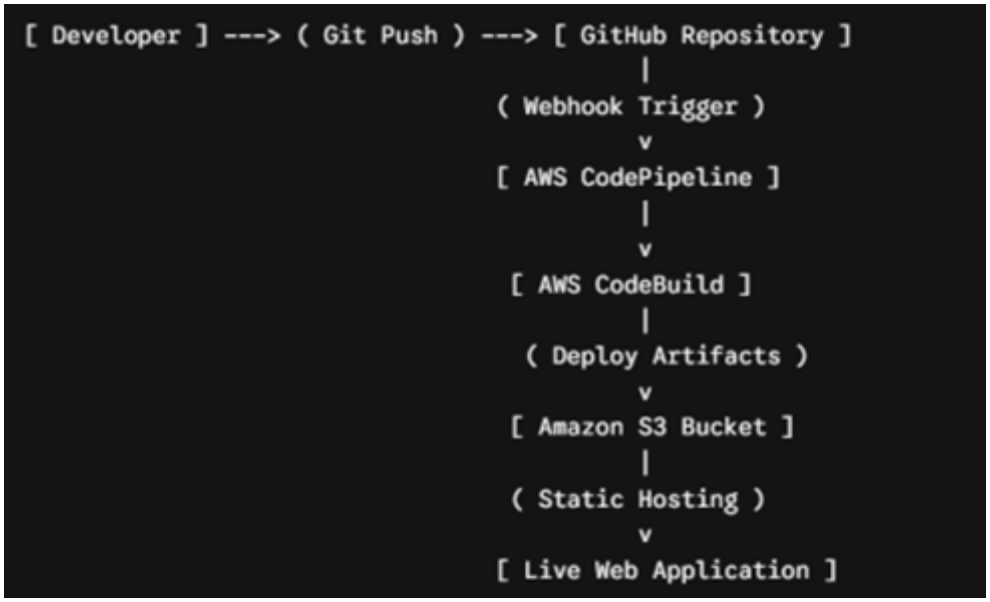
stages—such as Source, Build, and Deploy—moving build artifacts sequentially through execution pipelines.

- AWS CodeBuild: A fully managed, scalable build and compilation service. CodeBuild spins up isolated ephemeral compute containers, runs configuration scripts specified in a buildspec.yml file, runs unit tests, compiles static assets, and bundles output artifacts for downstream deployment.
- Amazon Simple Storage Service (Amazon S3): Provides scalable object storage with built-in Static Website Hosting. When configured with public read access (s3:GetObject) and an index.html root document, it serves as a lightweight, highly durable, serverless web server directly from bucket storage.
- AWS Identity and Access Management (IAM): Enforces fine-grained least-privilege security. Service roles ensure CodeBuild and CodePipeline have the exact permissions required (such as read/write access to S3 artifact and deployment buckets) to operate securely without exposing root or persistent user credentials.

Operational Workflow

- Push: A developer commits changes to the GitHub repository.
- Trigger: GitHub triggers an automated event notification to AWS CodePipeline via webhook integration.
- Build: CodePipeline retrieves the source files and triggers AWS CodeBuild to execute containerized compilation and packaging.
- Deploy: The pipeline extracts the verified build artifacts directly into the designated S3 deployment bucket.
- Serve: The Amazon S3 static website hosting endpoint renders the live web application immediately to users.

Architecture & Workflow:



Implementation:

Phase 1: AWS CodeBuild Project Configuration

Step 1: Create a Build Project

1. Open the AWS Management Console and navigate to Developer Tools > CodeBuild.
2. Click on Create build project.
3. Under Project configuration, set the Project name to pwa-codebuild-vb.
4. Select Default project as the project type.

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AWS account name
Choose a name for your account. You can change this name in your account settings after you sign up.

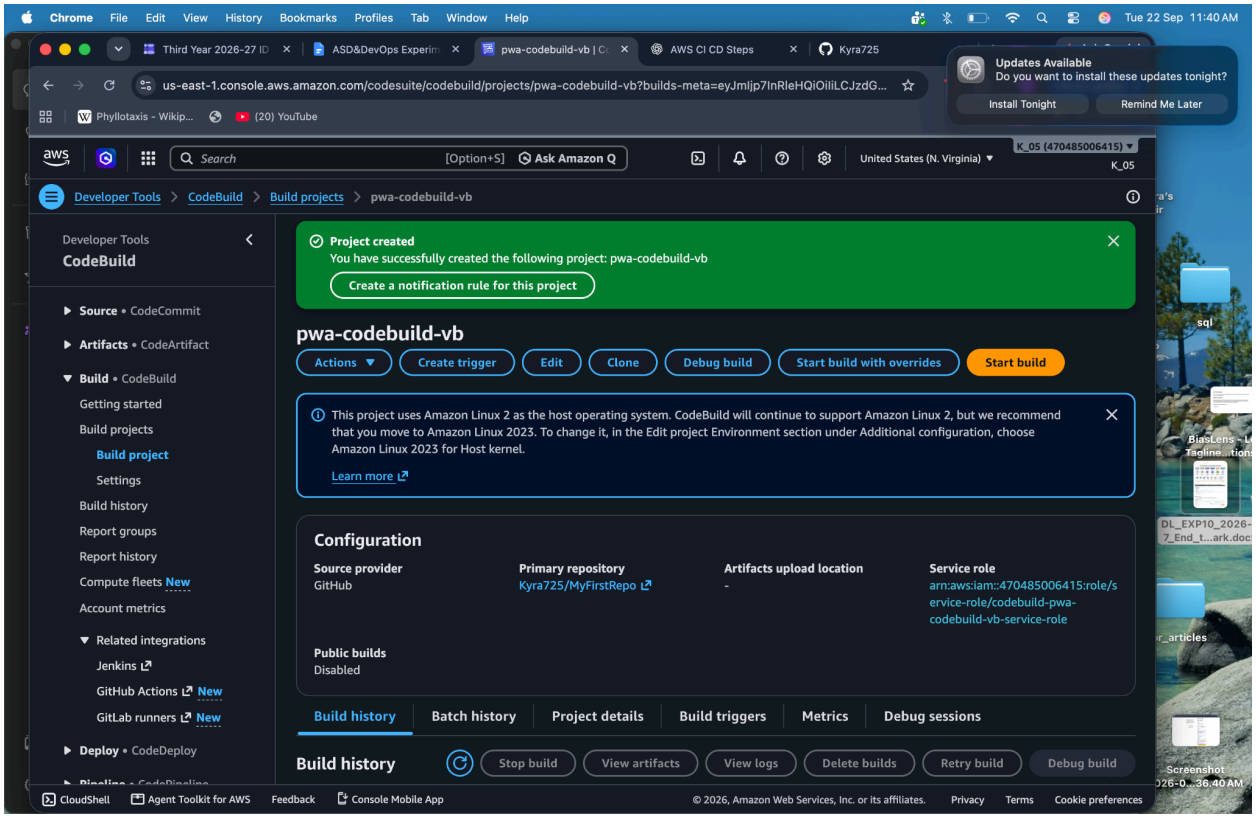
Irisha Deshpande

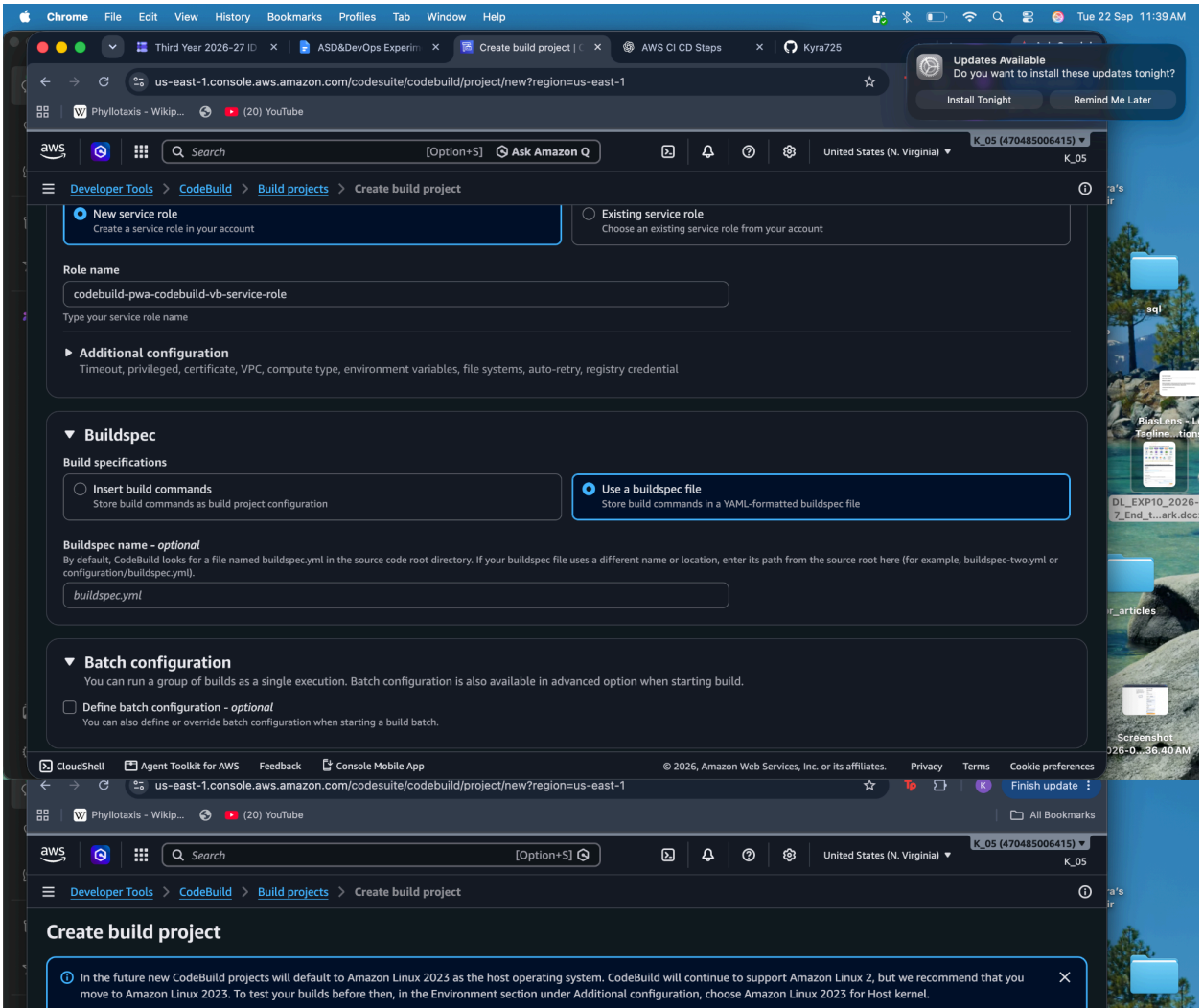
Verify email address

OR

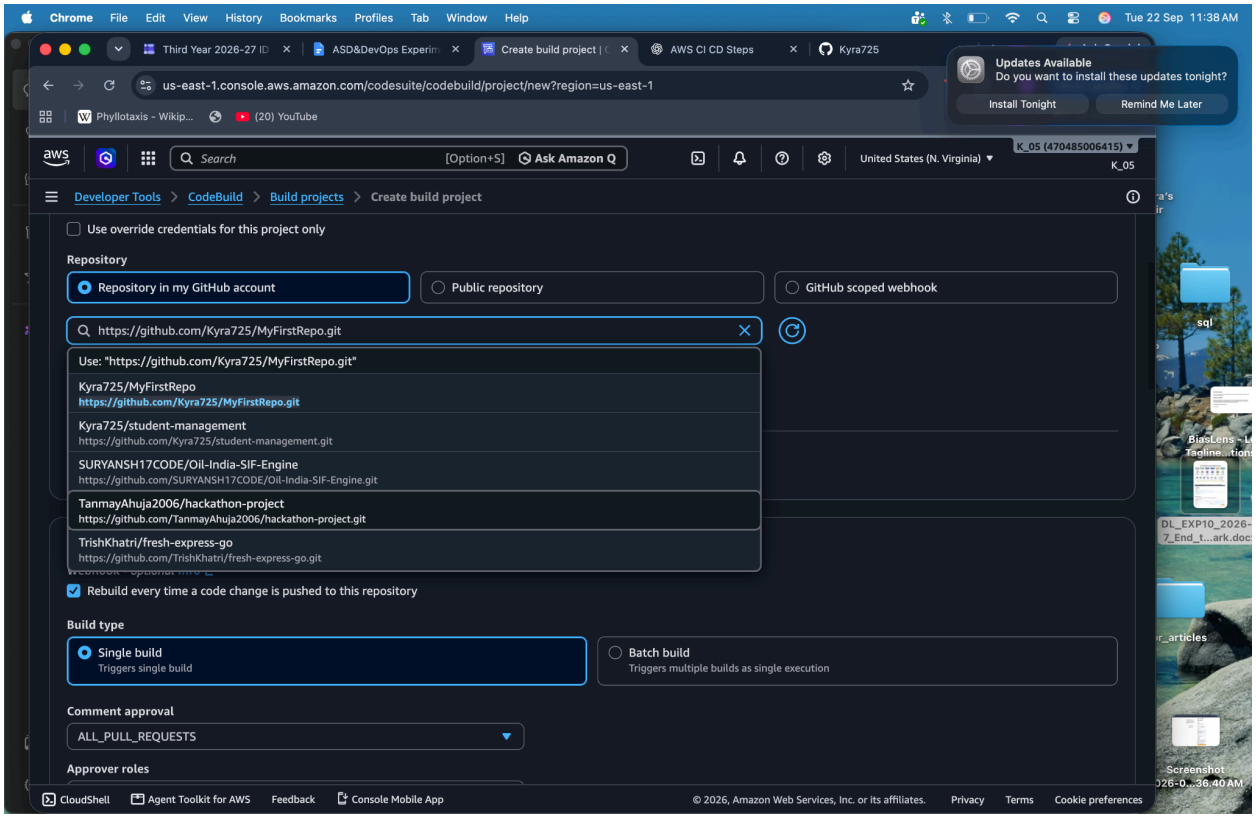
Sign in to an existing AWS account

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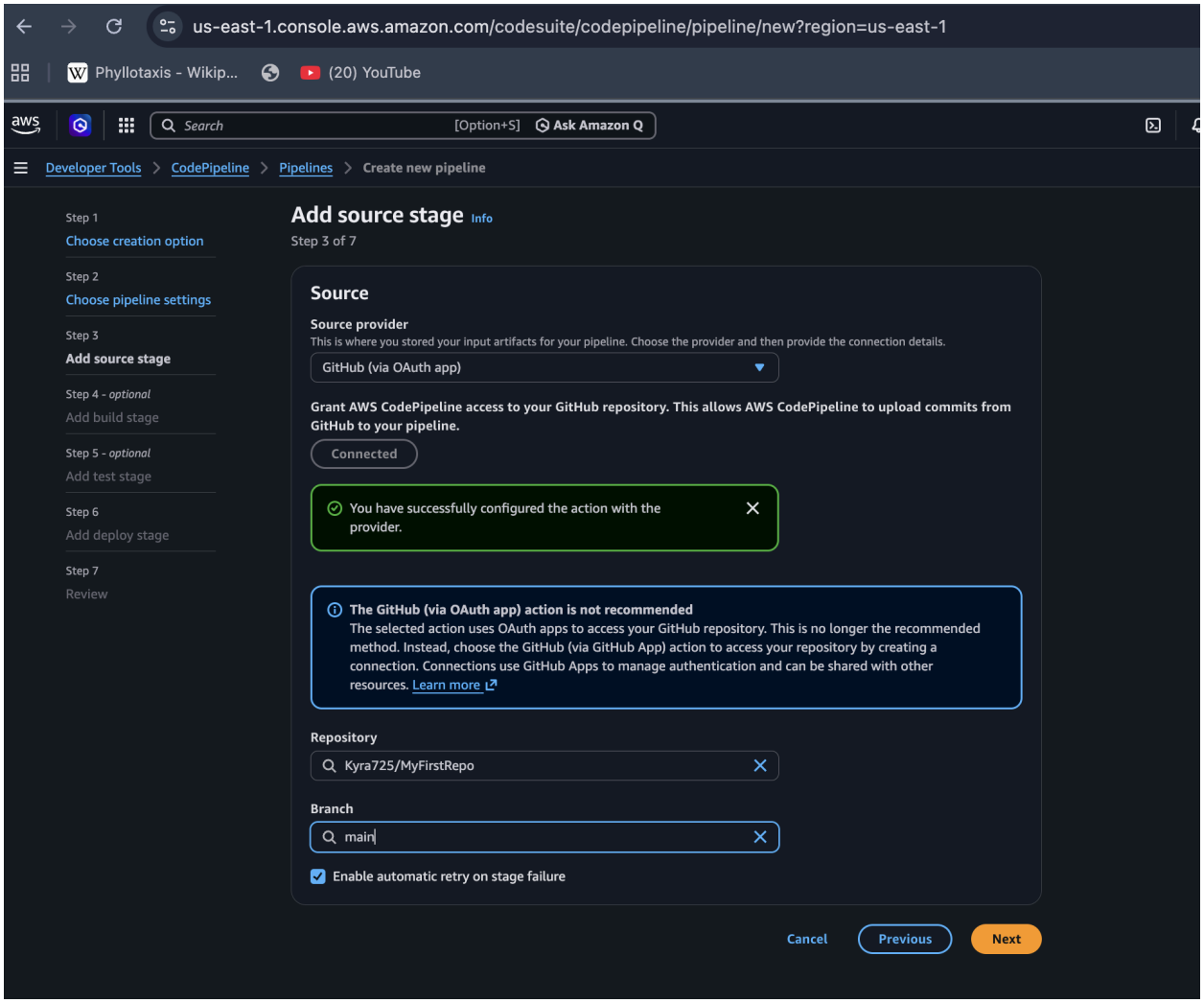

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Tue 22 Sep 11:40 AM
Updates Available
Do you want to install these updates tonight?
Install Tonight
Remind Me Later
us-east-1.console.aws.amazon.com/codesuite/codebuild/projects/pwa-codebuild-vb?builds-meta=eyJmIjp7InRleHQiOiIiLCJzdG...
Developer Tools > CodeBuild > Build projects > pwa-codebuild-vb
Project created
You have successfully created the following project: pwa-codebuild-vb
Create a notification rule for this project
pwa-codebuild-vb
Actions
Create trigger
Edit
Clone
Debug build
Start build with overrides
Start build
This project uses Amazon Linux 2 as the host operating system. CodeBuild will continue to support Amazon Linux 2, but we recommend that you move to Amazon Linux 2023. To change it, in the Edit project Environment section under Additional configuration, choose Amazon Linux 2023 for Host kernel.
Learn more
Configuration
Source provider
GitHub
Primary repository
Kyra725/MyFirstRepo
Artifacts upload location
-
Service role
arn:aws:iam::470485006415:role/service-role/codebuild-pwa-codebuild-vb-service-role
Public builds
Disabled
Build history
Batch history
Project details
Build triggers
Metrics
Debug sessions
Build history
Stop build
View artifacts
View logs
Delete builds
Retry build
Debug build
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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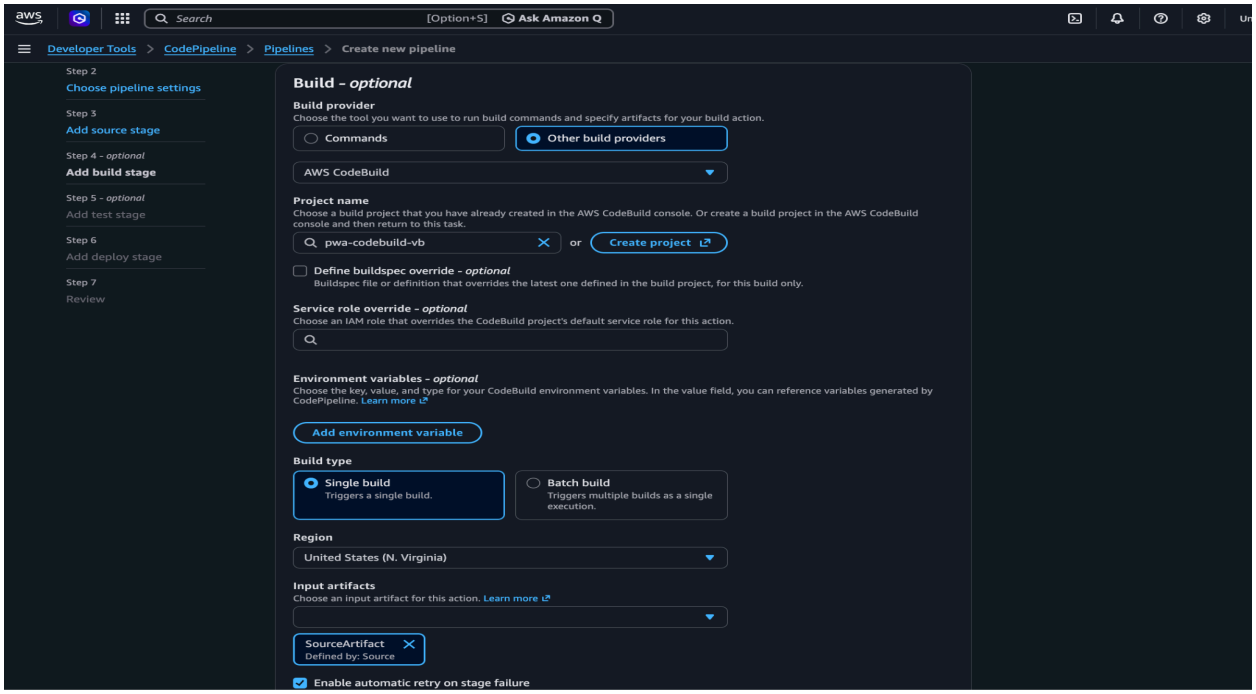

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Tue 22 Sep 11:39 AM
Updates Available
Do you want to install these updates tonight?
Install Tonight
Remind Me Later
us-east-1.console.aws.amazon.com/codesuite/codebuild/project/new?region=us-east-1
Developer Tools > CodeBuild > Build projects > Create build project
New service role
Create a service role in your account
Existing service role
Choose an existing service role from your account
Role name
codebuild-pwa-codebuild-vb-service-role
Type your service role name
Additional configuration
Timeout, privileged, certificate, VPC, compute type, environment variables, file systems, auto-retry, registry credential
Buildspec
Build specifications
Insert build commands
Store build commands as build project configuration
Use a buildspec file
Store build commands in a YAML-formatted buildspec file
Buildspec name - optional
By default, CodeBuild looks for a file named buildspec.yml in the source code root directory. If your buildspec file uses a different name or location, enter its path from the source root here (for example, buildspec-two.yml or configuration/buildspec.yml).
buildspec.yml
Batch configuration
You can run a group of builds as a single execution. Batch configuration is also available in advanced option when starting build.
Define batch configuration - optional
You can also define or override batch configuration when starting a build batch.
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Create build project
In the future new CodeBuild projects will default to Amazon Linux 2023 as the host operating system. CodeBuild will continue to support Amazon Linux 2, but we recommend that you move to Amazon Linux 2023. To test your builds before then, in the Environment section under Additional configuration, choose Amazon Linux 2023 for Host kernel.

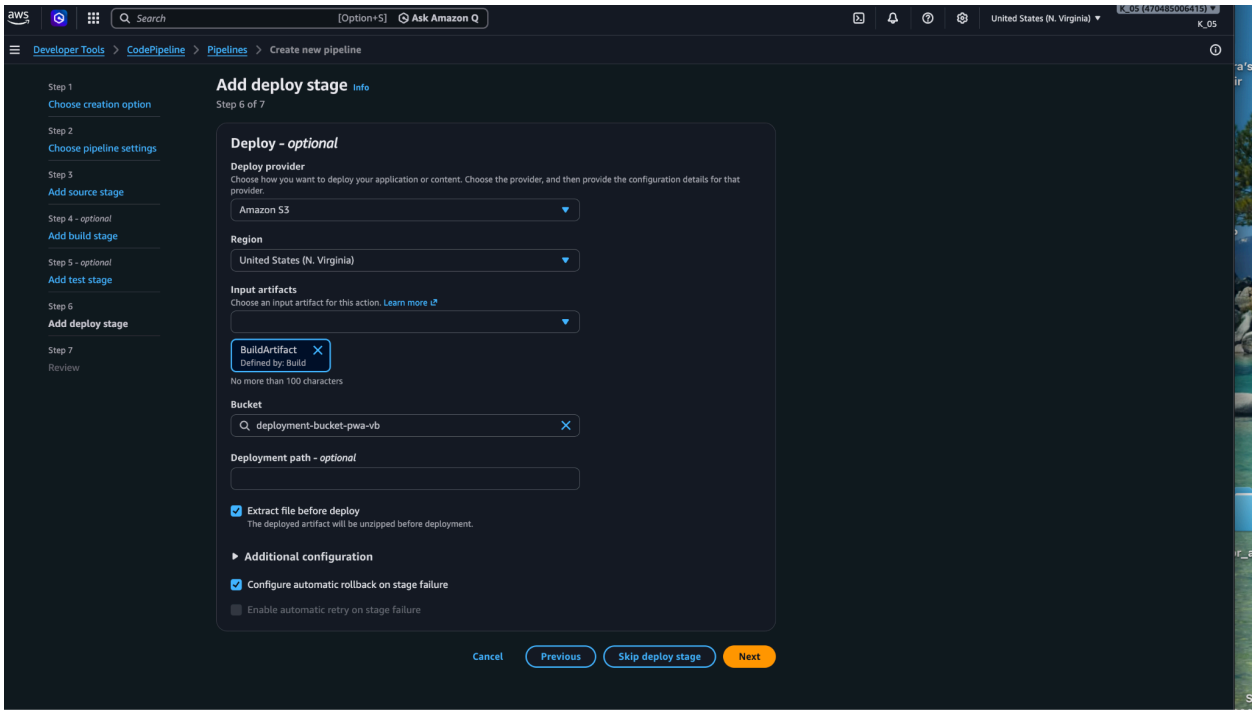
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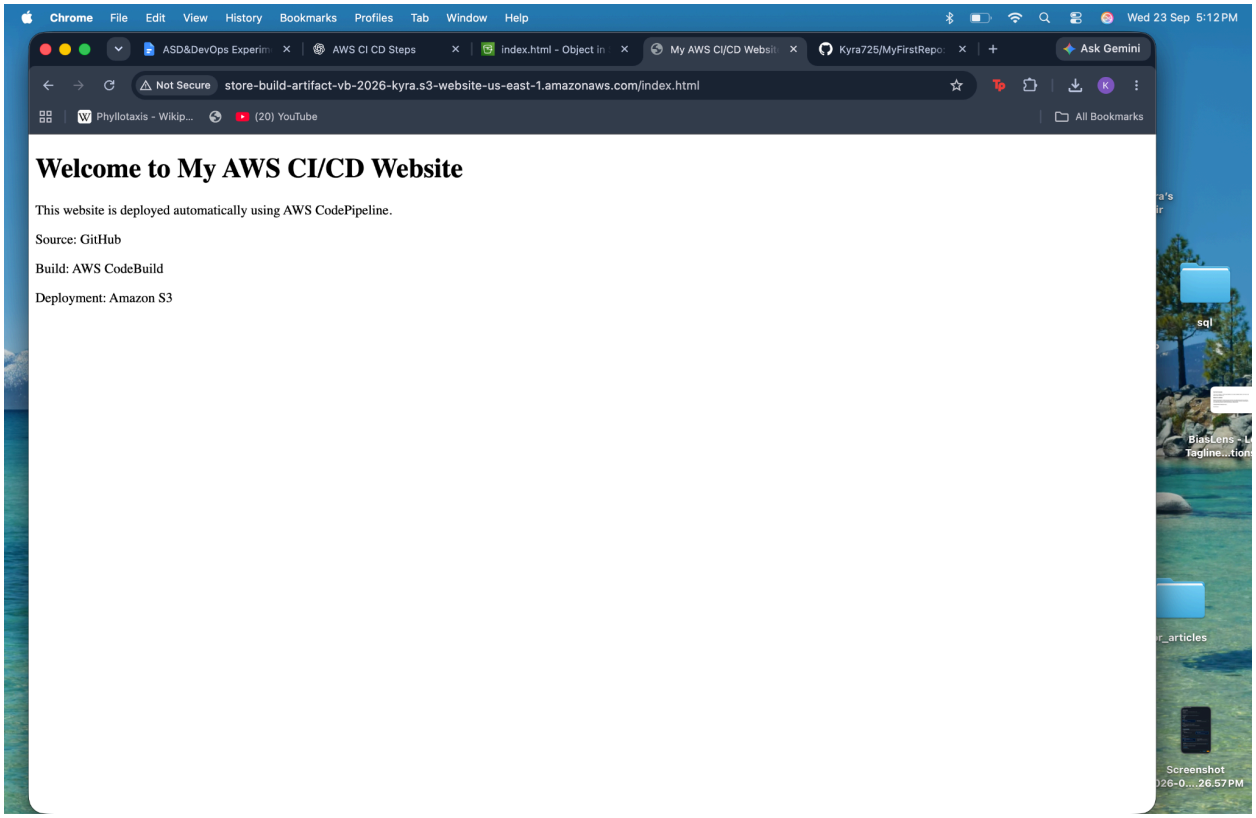


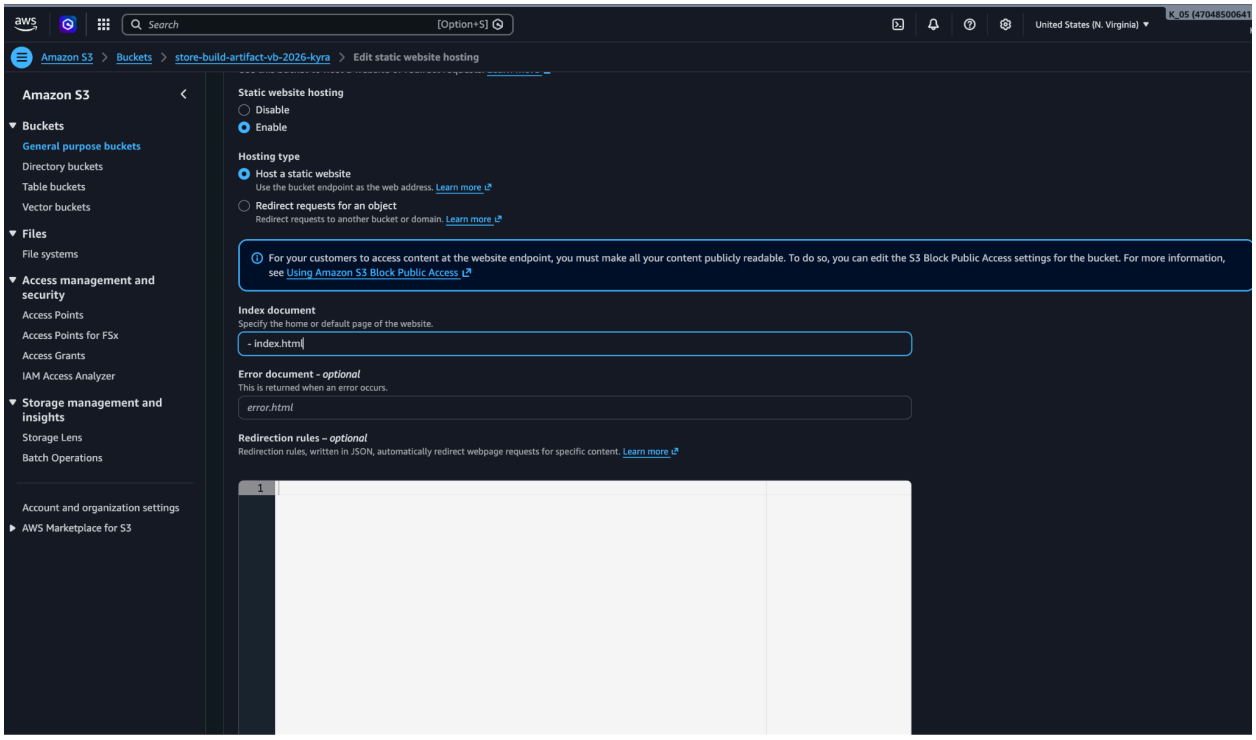

Choose pipeline settings Info
Step 2 of 7
Pipeline settings
Pipeline name
Enter the pipeline name. You cannot edit the pipeline name after it is created.
pwa-pipeline
No more than 100 characters
Execution mode Info
Choose the execution mode for your pipeline. This determines how the pipeline is run.
Superseded
Queued
Parallel
Service role
New service role
Create a service role in your account
Existing service role
Choose an existing service role from your account
Role name
AWSCodePipelineServiceRole-us-east-1-pwa-pipeline
Type your service role name
Allow AWS CodePipeline to create a service role so it can be used with this new pipeline
Advanced settings
Configure artifact store location, encryption settings, and pipeline variables for your pipeline.
Artifact store
Default location
Create a default S3 bucket in your account.
Custom location
Choose an existing S3 location from your account in the same region and account as your pipeline
Bucket
store-build-artifact-vb
Encryption key
Default AWS Managed Key
Use the AWS managed customer master key for CodePipeline in your account to encrypt the data in the artifact store.
Customer Managed Key
To encrypt the data in the artifact store under an AWS KMS customer managed key, specify the key ID, key ARN, or alias ARN.
Variables
You can add variables at the pipeline level. You can choose to assign the value when you start the pipeline. Learn more
No variables defined at the pipeline level in this pipeline.
Add variable
You can add up to 50 variables.
Cancel
Previous
Next


us-east-1.console.aws.amazon.com/codesuite/codepipeline/pipeline/new?region=us-east-1
Phyllotaxis - Wikip...
(20) YouTube
Search
[Option+S]
Ask Amazon Q
Developer Tools
CodePipeline
Pipelines
Create new pipeline
Step 1
Choose creation option
Step 2
Choose pipeline settings
Step 3
Add source stage
Step 4 - optional
Add build stage
Step 5 - optional
Add test stage
Step 6
Add deploy stage
Step 7
Review
Add source stage
Info
Step 3 of 7
Source
Source provider
This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.
GitHub (via OAuth app)
Grant AWS CodePipeline access to your GitHub repository. This allows AWS CodePipeline to upload commits from GitHub to your pipeline.
Connected
You have successfully configured the action with the provider.
The GitHub (via OAuth app) action is not recommended
The selected action uses OAuth apps to access your GitHub repository. This is no longer the recommended method. Instead, choose the GitHub (via GitHub App) action to access your repository by creating a connection. Connections use GitHub Apps to manage authentication and can be shared with other resources. Learn more
Repository
Kyra725/MyFirstRepo
Branch
main
Enable automatic retry on stage failure
Cancel
Previous
Next

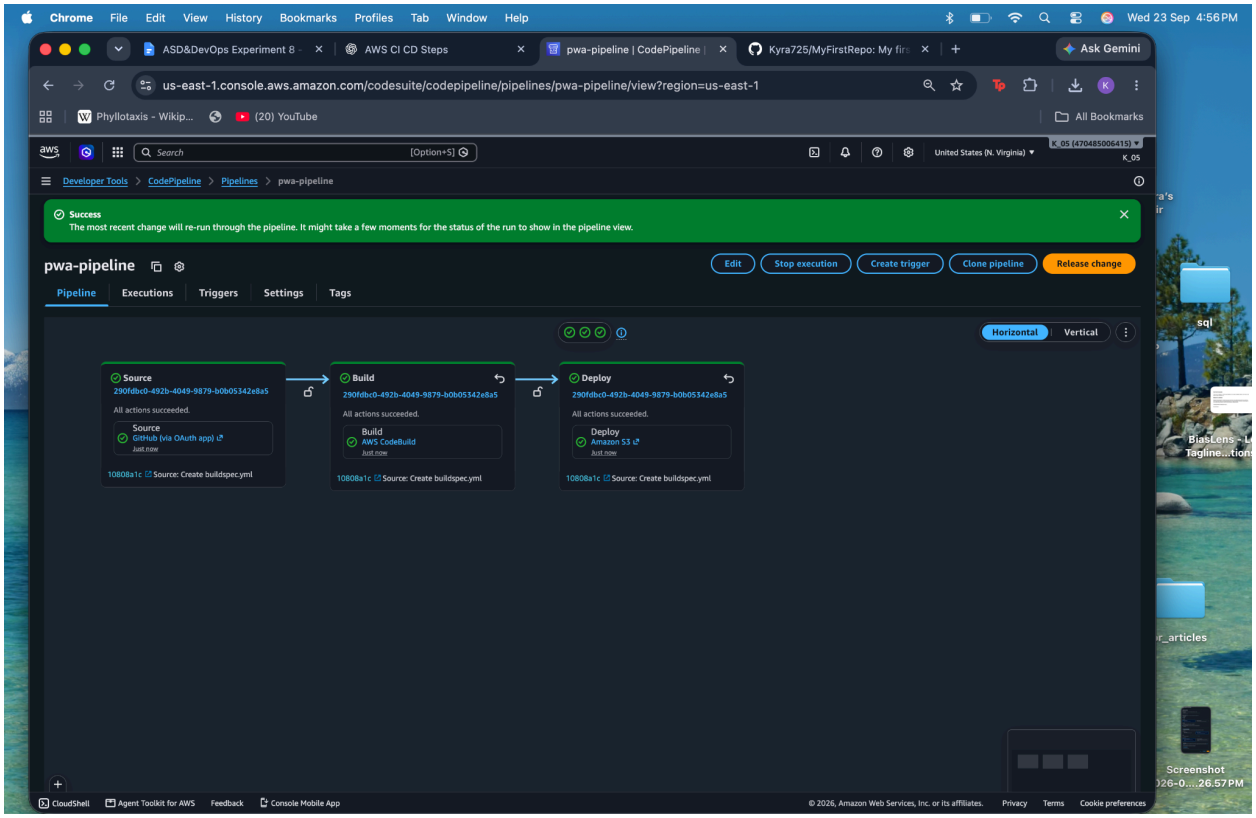

Search
[Option+S]
Ask Amazon Q
Developer Tools > CodePipeline > Pipelines > Create new pipeline
Step 2
Choose pipeline settings
Step 3
Add source stage
Step 4 - optional
Add build stage
Step 5 - optional
Add test stage
Step 6
Add deploy stage
Step 7
Review
Build - optional
Build provider
Choose the tool you want to use to run build commands and specify artifacts for your build action.
Commands
Other build providers
AWS CodeBuild
Project name
Choose a build project that you have already created in the AWS CodeBuild console. Or create a build project in the AWS CodeBuild console and then return to this task.
pwa-codebuild-vb
or
Create project
Define buildspec override - optional
Buildspec file or definition that overrides the latest one defined in the build project, for this build only.
Service role override - optional
Choose an IAM role that overrides the CodeBuild project's default service role for this action.
Environment variables - optional
Choose the key, value, and type for your CodeBuild environment variables. In the value field, you can reference variables generated by CodePipeline. Learn more
Add environment variable
Build type
Single build
Triggers a single build.
Batch build
Triggers multiple builds as a single execution.
Region
United States (N. Virginia)
Input artifacts
Choose an input artifact for this action. Learn more
SourceArtifact
Defined by: Source
Enable automatic retry on stage failure

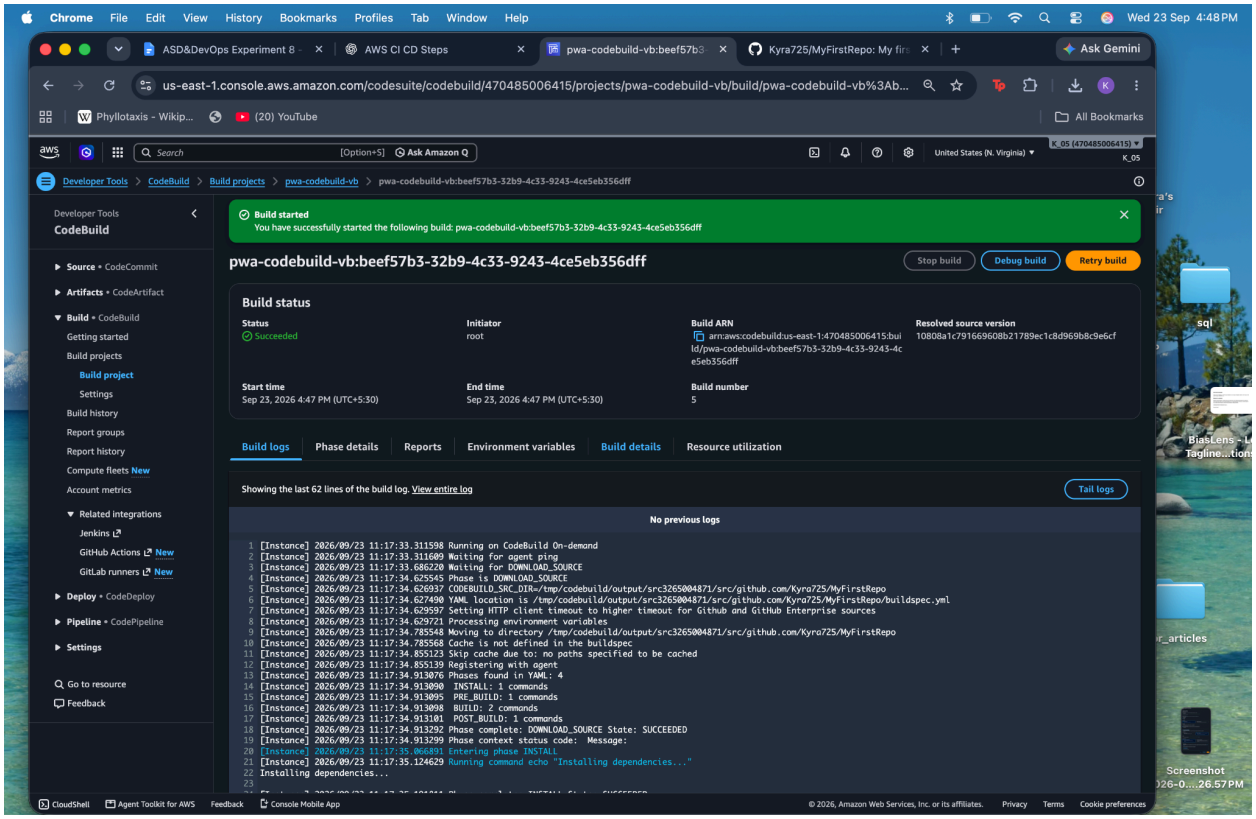

Search
[Option+S]
Ask Amazon Q
United States (N. Virginia)
Developer Tools > CodePipeline > Pipelines > Create new pipeline
Step 1
Choose creation option
Step 2
Choose pipeline settings
Step 3
Add source stage
Step 4 - optional
Add build stage
Step 5 - optional
Add test stage
Step 6
Add deploy stage
Step 7
Review
Add deploy stage Info
Step 6 of 7
Deploy - optional
Deploy provider
Choose how you want to deploy your application or content. Choose the provider, and then provide the configuration details for that provider.
Amazon S3
Region
United States (N. Virginia)
Input artifacts
Choose an input artifact for this action. Learn more
BuildArtifact
Defined by: Build
No more than 100 characters
Bucket
deployment-bucket-pwa-vb
Deployment path - optional
Extract file before deploy
The deployed artifact will be unzipped before deployment.
Additional configuration
Configure automatic rollback on stage failure
Enable automatic retry on stage failure
Cancel
Previous
Skip deploy stage
Next

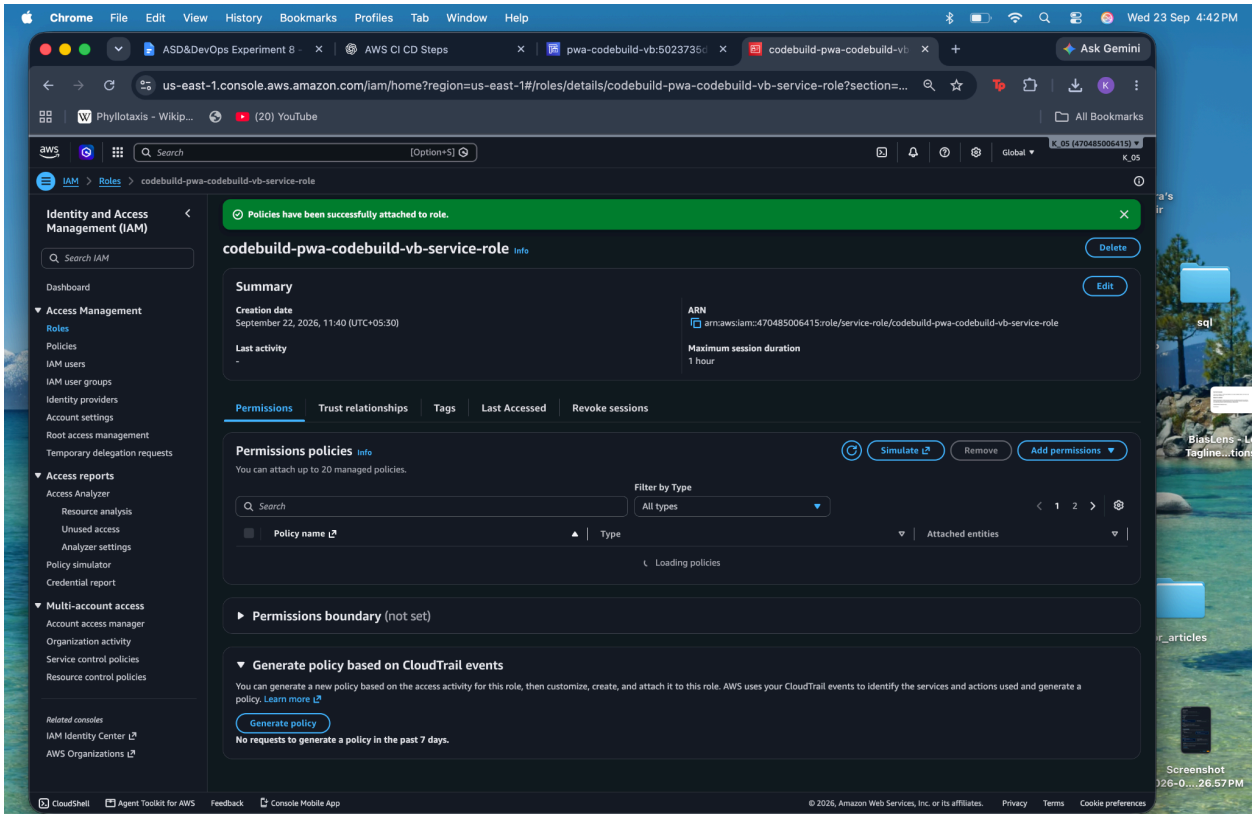

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Wed 23 Sep 5:12 PM
Not Secure
store-build-artifact-vb-2026-kyra.s3-website-us-east-1.amazonaws.com/index.html
Welcome to My AWS CI/CD Website
This website is deployed automatically using AWS CodePipeline.
Source: GitHub
Build: AWS CodeBuild
Deployment: Amazon S3

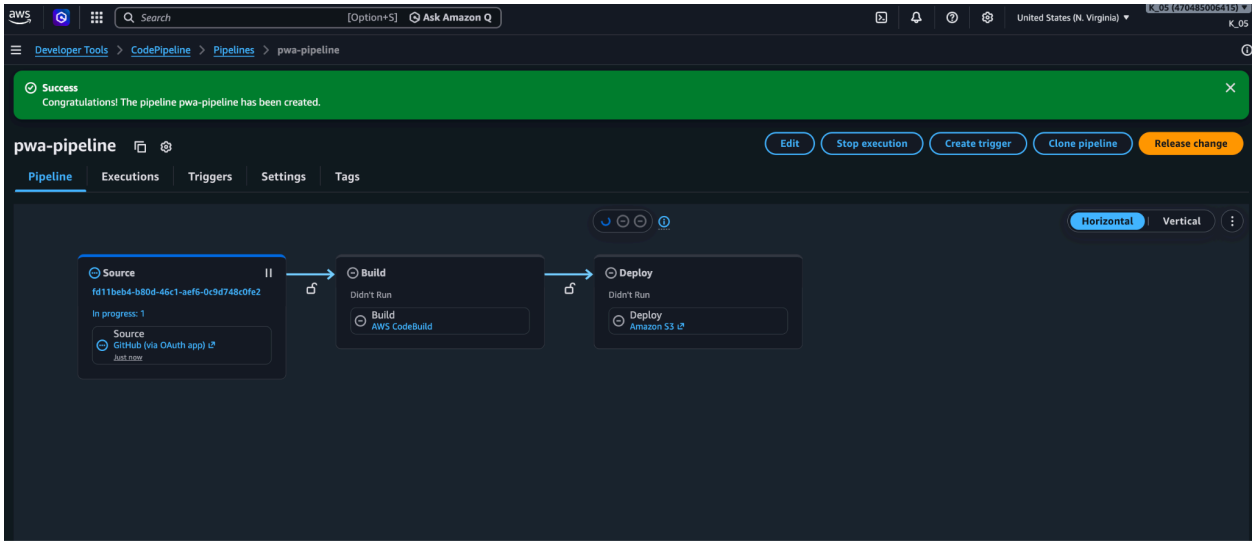

Amazon S3 > Buckets > store-build-artifact-vb-2026-kyra > Edit static website hosting
Static website hosting
Disable
Enable
Hosting type
Host a static website
Redirect requests for an object
Index document
- index.html
Error document - optional
error.html
Redirection rules – optional

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Wed 23 Sep 4:42 PM
us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/roles/details/codebuild-pwa-codebuild-vb-service-role?section=...
IAM > Roles > codebuild-pwa-codebuild-vb-service-role
Identity and Access Management (IAM)
Policies have been successfully attached to role.
codebuild-pwa-codebuild-vb-service-role Info
Delete
Summary
Edit
Creation date
September 22, 2026, 11:40 (UTC+05:30)
ARN
arn:aws:iam::470485006415:role/service-role/codebuild-pwa-codebuild-vb-service-role
Last activity
-
Maximum session duration
1 hour
Permissions
Trust relationships
Tags
Last Accessed
Revoke sessions
Permissions policies Info
You can attach up to 20 managed policies.
Simulate
Remove
Add permissions
Filter by Type
All types
Policy name
Type
Attached entities
Loading policies
Permissions boundary (not set)
Generate policy based on CloudTrail events
You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. Learn more
Generate policy
No requests to generate a policy in the past 7 days.
Dashboard
Access Management
Roles
Policies
IAM users
IAM user groups
Identity providers
Account settings
Root access management
Temporary delegation requests
Access reports
Access Analyzer
Resource analysis
Unused access
Analyzer settings
Policy simulator
Credential report
Multi-account access
Account access manager
Organization activity
Service control policies
Resource control policies
IAM Identity Center
AWS Organizations
CloudShell
Feedback
Console Mobile App
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Developer Tools > CodePipeline > Pipelines > pwa-pipeline
Success
Congratulations! The pipeline pwa-pipeline has been created.
pwa-pipeline
Edit
Stop execution
Create trigger
Clone pipeline
Release change
Pipeline
Executions
Triggers
Settings
Tags
Horizontal
Vertical
Source
fd11beb4-b80d-46c1-aef6-0c9d748c0fe2
In progress: 1
Source
GitHub (via OAuth app)
Just now
Build
Didn't Run
Build
AWS CodeBuild
Deploy
Didn't Run
Deploy
Amazon S3

Conclusion:

LO mapped successfully.